DMBI

Aprior Algorithm

Q. Use Aprior algo. to identify Item set ...... Assume Support 50%. Min Confi. 75%.

Sol:

Tid	items
a	1, 2, 4, 5, 6
b	2, 3, 5
c	1, 2, 4, 5
d	1, 2, 4, 5
e	1, 2, 3, 4, 5, 6
f	2, 3, 4
g	1, 2, 4, 5

Step1: Find Support for itemsets (L_1)

item	Count	Support	Status (Min 50%)
1	5/7	71.4 %	Accept
2	7/7	100%	Accept
3	3/7	42%	Reject
4	6/7	85%	Accept
5	6/7	85%	Accept
6	2/7	28.5%	Reject

Item remaining: { 1, 2, 4, 5 }

Step 2: find Support for itemsets (L_2)

items	Sup Count	Support	Status (Min50)
{1, 2}	5/7	71.4%	Accept
{1, 4}	5/7	71.4%	
{1, 5}	5/7	71.4%	
{2, 4}	6/7	85.7%	
{2, 5}	6/7		
{4, 5}	5/7		

Step3 : find Support for itemsets (L_3)

{1, 2, 4}	5/7	71.4	Accept
{1, 2, 5}	5/7		
{1, 4, 5}	5/7		
{2, 4, 5}	5/7		

Step4: L_4

{1, 2, 4, 5}	5	71.4	Accpt

Final Itemset: {1, 2, 4, 5}

Now, we calculate confidence from final itemset

$$\text{Confidence } (A \Rightarrow B) = \frac{\text{Support } (A \cup B)}{\text{Support } A}$$

Numerator → final ka total count
denominator → Left ka count

Rule1

$\{1,2,4\} \Rightarrow 5$ 83.3%
Confidence $5/6$ = strong

Rule2

$\{1,4,5\} \Rightarrow 2$
Confidence $5/5$ 100%

Rule3

$2 \Rightarrow \{1,4,5\}$
Confidence $5/7$ = 71.4% (Reject

Final Result:

Final itemset $\{1,2,4,5\}$
Strong Rule $\{1,4,5\} \Rightarrow 2$ with 100% confidence

Note: We have pairs of two
{AB}, {AD}, {AE}, {BC}, {BD}, {BE}
{CE}, {DE}
Now, make pairs of three sets
i) Choose first letter same as prefix
{AB} + {AD} ⇒ {ABD} ✓
{AB} + {AE} ⇒ {ABE} ✓
{AD} + {AE} ⇒ {ADE} ✓ Same for B.
{CE} Reject bcz ek hi hai
Three se four → No match
{ABD}, {ABE} {ADE} {BCE} {BDE}
{ABDE} {A, B, D E} {B, C, D E}
(i) ↓ (ii)
Repeat
ho gya

Naive Bayes Algorithm

Q. Use Naive bayes Algo with hot temprerature, rainy outlook, high humidity & no wind

outlook	temp.	Humidity	Windy	Class
Sunny	hot	high	false	NO
Sunny	hot	high	true	NO
overcast	hot	high	False	Play
Rain	mild	high	false	
rain	cool	normal	False	
rain			T	NO
overcast			T	Play
Sunny		high	F	NO
Sunny		normal	F	Play
rain			F	
Sunny			T	
overcast		high	T	
overcast	hot	normal	F	
rain	mild	high	T	NO

eg.- x = (outlook = rain, Temperature = hot, humidity = high, Wind = Flase)
Using Naivy bayes on class play tennis dataset

Step1: Prior probabilities
total records = 14
Play = 9
No = 5

$P(Play) = \frac{9}{14}$, $P(NO) = \frac{5}{14}$

Step 1 Give possibilities
count among Play:

$P(rain|Play) = 3/9$
$P(hot|play) = 2/9$
$P(high|Play) = 3/9$
$P(false|play) = 6/9$

$$P(X|Play) = \frac{3}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{6}{9}$$

Step 2: Given possibilities
count among no:

$P(rain|NO) = 2/5$
$P(hot|NO) = 2/5$
$P(high|NO) = 4/5$
$P(false|NO) = 2/5$

$$P(X|NO) = \frac{2}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{2}{5}$$

Step 3: Proportional

For Play $P(X|Play) \propto \frac{9}{14} \times \frac{3}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{6}{9} = 0.0105$

For NO $P(X|NO) \propto \frac{5}{14} \times \frac{2}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{2}{5} = 0.0183$

Step 4: Comparison

$0.0183 > 0.0105$

Final decision: class = NO

K-Mean clustering

Q. Explain steps in hierarchical cluster algo. Perform clustering data (2,3), (5,4), (9,6), (4,7), (8,1), (7,2), (6,3), (1,9), (3,6) (4,8) consider 3 cluster...

Sol. Step 1:
Chose initial cluster/centroids
$C_1(2,3)$, $C_2(5,4)$, $C_3(9,6)$

Step 2:
Assign points to nearest centroid
using $D=(x_2-x_1)^2+(y_2-y_1)^2$

Points (x,y)	Dist to $(C_1)^2$	(Dist to $C_2)^2$	(Dist to $C_3)^2$	Nearest cluster
(2,3)	0	10	58	C_1
(5,4)	10	0	20	C_2
(9,6)	58	20	0	C_3
(4,7)	20	10	26	C_2
(8,1)	40	18	26	C_2
(7,2)	26	8	20	C_2
(6,3)	16	2	18	C_2
(1,9)	37	41	73	C_1
(3,6)	10	8	36	C_2
(4,8)	29	17	29	C_2

Step3: Update Centroids

Cluster1 : (2,3) (1,9)

$x = \frac{2+1}{2} = 1.5, \quad y = 3+9/_2 = 6$

New $C_1 = (1.5, 6)$

Cluste2: (5,4), (4,7) (8,1), (7,2), (6,3), (3,6) (4,8)

$x = 5+4+8+7+6+3+4/_7 = 5.28$

$y = 4.43$

New $C_2 = 5.28, 4.43$

&

New $C_3 = (9,6)$ ∴ (only one point)

Step4: final Conclusion
updated cluster after iteration1:

Cluster 1 : {(2,3), (1,9)}
cluster 2 : {(5,4), (4,7), (8,1), (7,2), (6,3), (3,6) (4,8)}
cluster 3 : {9,6}

updated Centroids
$C_1 (1.5, 6)$
$C_2 (5.28, 4.43)$
&
$C_3 (9,6)$

Min-Max Algorithm & Z-Score Normalization

Q. Given dataset [10, 20, 30, 40, 50]. Apply min-max normalization (Range [0, 1] also apply z-score normalization for input 30.

Sol. Min max Normalization to range [0,1]

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Here,

$x_{min} = 10$, $x_{max} = 50$

so,

$$x' = \frac{x - 10}{40}$$

Now,

x	Normalized x'
10	0.00
20	0.25
30	0.50
40	0.75
50	1.00

Z-Score normalization for input 30

formula:

$$Z = \frac{x - \mu(\text{mean})}{\sigma\ (SD)}$$

ye 30 mean wala hai

Here, $\mu = 10 + 20 + 30 + 40 + 50 / 5 = 30$

& $(\sigma)SD = \sqrt{(10-30)^2 + (20-30)^2 + (30-30)^2 + (40-30)^2 + (50-30)^2 / 5}$

$= \sqrt{200}$

$= 14.14$

Now,

Z-Score for $x = 30$

$z = 30 - 30 / 14.14 = 0$

ye Question me given tha.

Note:

z-score given 50 for this question

$z = 50 - 30 / 14.14 = 1.41$

final Conclusion

Min-max normalized Value: 0, 0.25, 0.75, 1

Z score of 30: 0

Performance Evaluation (confusion matrix)

TP (Actual Yes, predict Yes)
↳ True positive

FN (Actual yes, predict NO)
↳ False ~~positive~~ negative

FP (false positive)
↳ Actual no, predict yes)

TN (Actual NO, Predict NO)
↳ True Negative

Q. Give confusion matrix, Calculate Accuracy & Precision.

	Predicted Yes	Predicted NO
Actual yes	50	10
Actual NO	5	35

Sol. Accuracy :

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{50 + 35}{50 + 35 + 5 + 10} = \frac{85}{100} = 0.85$$

Accuray = 85%

Precision :

$$\text{precision} = \frac{TP}{TP + FP}$$

$$= \frac{50}{50+5} = \frac{50}{55}$$

$$= 0.909$$

precision = 90.9%

Simple Leakage Algorithm
&
Smoothing Bin Mean, Bin Boundary ....

Q. 21

Object	x	y
A	1	1
B	1.5	1.5
C	5	5
D	3	4
E	4	4
F	3	3.5

Apply Single leakage Clustering & draw dendrogram

Step1: Find distance b/w each & every Point using $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

	A	B	C	D	E	F
A	0	0.70	5.65	3.60	4.24	3.20
B		0	4.94	2.91	3.53	2.5
C			0	2.23	1.41	2.5
D				0	1	0.5
E					0	1.11
F						0

Step2: i) Find min. dist & merge cluster

	A	B	C	DF	E
A	0	0.70	5.65	3.60	4.24
B		0	4.94	2.91	3.53
C			0	2.23	1.41
DF				0	1
E					0

ii)

	AB	C	DF	E
AB	0	5.65	3.60	4.24
C		0	2.23	1.41
DF			0	1
E				0

iii)

	AB	C	DFE
AB	0	5.65	3.60
C		0	2.23
DFE			0

iv)

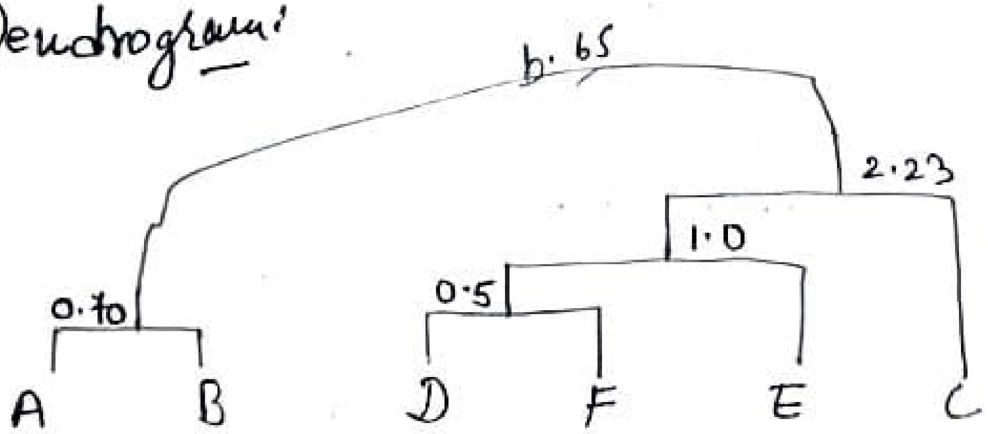
	AB	DFEC
AB	0	5.65
DFEC		0

v) Now merge AB DFEC

Final cluster Sequence:-

(D,F) → (A, B) → (D,F) E → ((D,F),E) C → (A,B) (D,E,

Dendrogram:



Q.2 (i) Sol.

Smoothing binwala Question h Yehh

Step 1: Arrange in Ascending Order

Data: 11, 13, 13, 15, 15, 16, 19, 20, 20, 20, 21, 21, 22,
23, 24, 30, 40, 45, 45, 45, 71, 72, 73, 75

Total = 24
observation

Given Bins = 4

So,

size of one Bin = 24/6 = 6

Step 2: Distributed the data into four Bins

Bin1: 11, 13, 13, 15, 15, 16
Bin2: 19, 20, 20, 20, 21, 21
Bin3: 22, 23, 24, 30, 40, 45
Bin4: 45, 45, 71, 72, 73, 75

Step 3: Smoothing Bins Mean

Bin1 (Mean) = (11+13+13+15+15+16)/6 = 13.83 ≈ 14
So, smoothed Bin = 14, 14, 14, 14, 14, 14
Bin2 (Mean) = (19+20+20+20+21+21)/6 = 20
So, Smoothed Bin = 20, 20, 20, 20, 20, 20
Bin3 (Mean) = (22+23+24+30+40+45)/6
= 30.66 ≈ 31

Smoothed Bin = 31, 31, 31, 31, 31, 31

Bin4 = (45+45+71+72+73+75)/6
= 63.5 ≈ 64

Smooter Bin → 64, 64, 64, 64, 64

Step4 smoothing Bin Meadian

Bin1 Meadian : (13+15)/2 = 14 so, 14, 14, 14, 14, 14, 1
Bin2 Median ! (20+20)/2 = 20 so 20, 20, 20, 20, 20, 2
Bin3 Median : (24+30)/2 = 27, 27, 27, 27, 27, 27
Bin4 Median : (71+72)/2 = 71.50

Step5 Smoothed data By Boundary

Bin1 (11 - 16) : 11, 11, 11, 16, 16, 16.
Bin2 (19 - 21) : 19, 19, 19, 21, 21, 21
Bin3 (22 - 45) : 22, 22, 22, ~~45~~ 22, 45, 45
Bin4 (45 - 75) : 45, 45, 75, 75, 75, 75

Q.2 (2) Sol. (i) Mean = $\frac{\text{Sum of data}}{\text{Total observation}}$

= (13+15+16+16+19+20+21
+22+22+22+25+25+25+25
+30+33+33+35+35+35
+35+36+40+45+
46+52+70) / 26 = 789 / 26

Mean = 30.35

Middle/2

Median = 25+30/2 = 27.5

ii) Mode = 25 & 35 (appears both 4 times)
(Two data so its Bimodal)

iii) Mid Range = $\frac{Min+Max}{2} = \frac{13+70}{2} = 41.5$

iv) five point Summary

~~Minimum Value = 13~~
~~Maximum Value = 70~~ → last # Likho!
~~Mean = 30.35~~
~~Median = 27.5~~
~~Mind = 41.5~~

ORder(Min → Q1 → Q2 (Median) → Q3 → Max

Minimum = 13
Q1 → lower half
13, 15, 16, 16, 19, 20, 21, 22, 22, 25, 25, 25, 25
Middle of this (7th value) = 21
Q1 = 21

Q3 → upper half → 35

Max 70

So, 13, 21, 27.5, 35, 70

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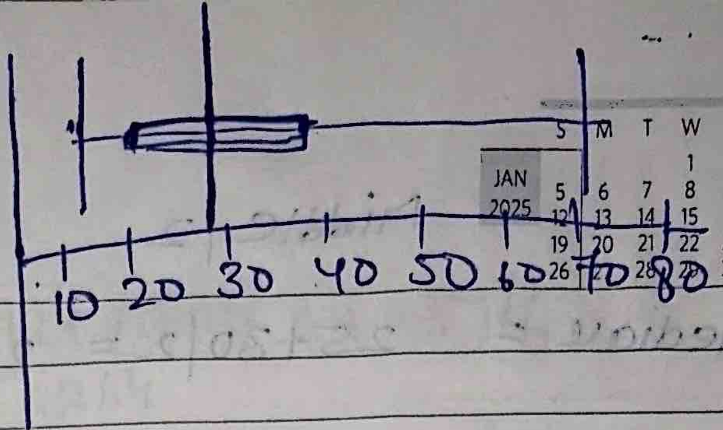
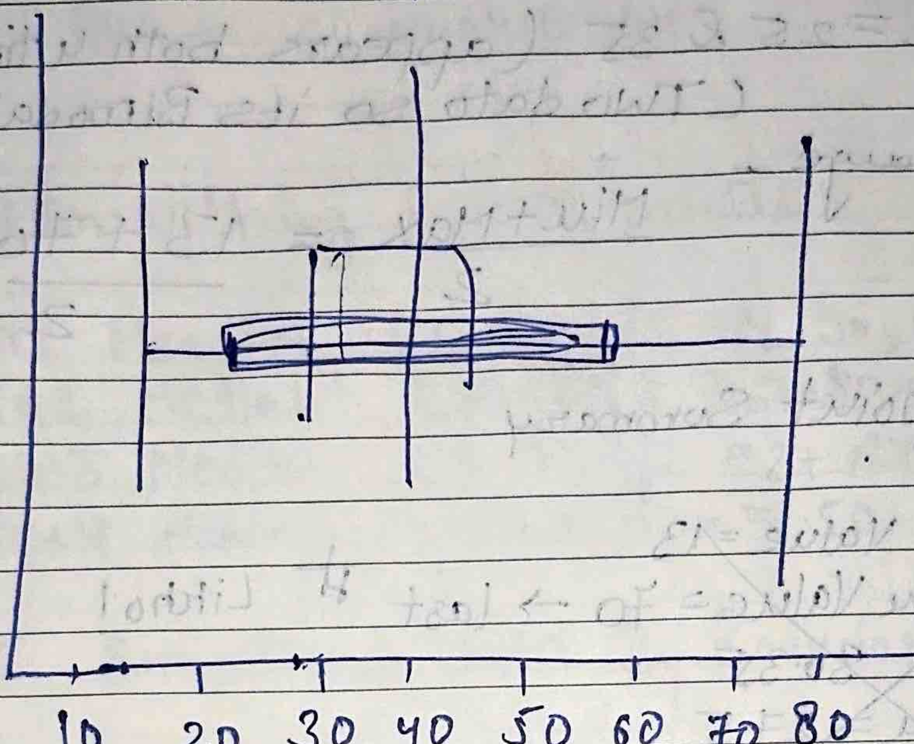
S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	



v) Graph



Min → Q1 → Median → Q3 → Max

13, 21, 27.5, 35, 70